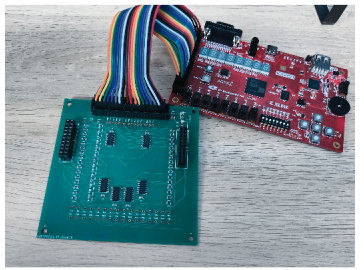
XUNJI

Intelligent Wiring Harness Topology Recognition and Dynamic Fault Diagnosis Using FPGA Parallel Timing Analysis

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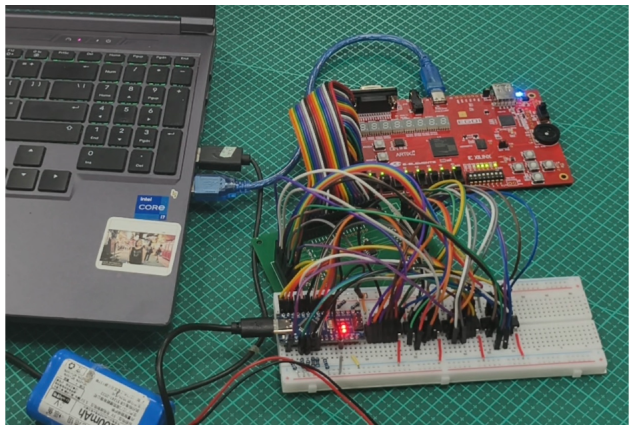
China University of Geosciences (Wuhan)



EGO1 / AMD Artix-7 XC7A35T

INTRODUCTION

Miswiring, open circuits and intermittent contacts affect equipment reliability. XUNJI combines 16-channel topology reconstruction, fault localization and transient-event recording in one FPGA-based prototype.



Prototype setup with EGO1, FE-16 and Nano/CD4066 fault injection

APPLICATIONS

Assembly checks: detect missing, crossed and bridged connections.
Maintenance: locate specific A-to-B connection differences.
Reliability tests: retain disconnection and recovery events.

16 channels, 256 candidate connections

SYSTEM ARCHITECTURE

FE-16 excitation and signal conditioning

↓

FPGA topology reconstruction and diagnosis

↓

Snapshot / UART / CRC-16

↓

Host visualization and event recording

DESIGN

Learnable topology diagnosis

Learn a normal reference matrix. Compare missing and extra links to distinguish open circuits, miswiring, bridges and compound anomalies.

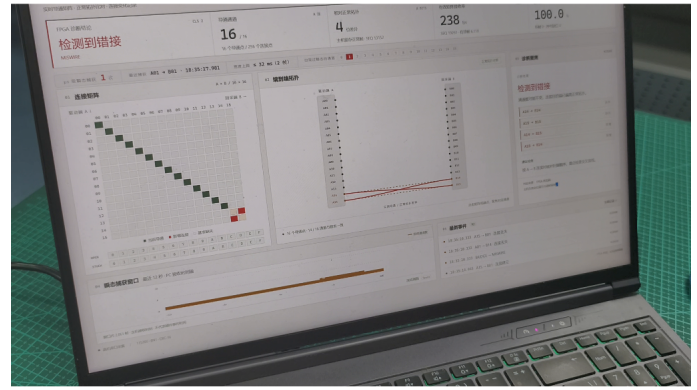
Microsecond transient capture

Use a 100 MHz hardware timebase for sampling and event detection. Share the front end between topology and transient observation.

Decoupled scan and transmission

Latch a matrix snapshot before transmission so the payload and CRC use identical data while scanning continues.

RESULTS



Real miswiring diagnosis: matrix and crossed links

Microsecond transient capture

Channels / matrix	**16 / 16 × 16**
Clock	**100 MHz**
LUT	**2119 (10.19%)**
FF	**1564 (3.76%)**
BRAM / DSP	**0 / 0**
WNS	**+1.408 ns**

Validation: 6 RTL testbenches, 84 pytest checks

Resources, timing and tests: supplied report (16 channels).